Empirical ERDS Characterization — Motor Imagery

Generated mi_erds_20260806T235629Z. Dataset: **PhysioNet EEGMMIDB** (motor imagery, runs [4, 8, 12]), subjects **[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]** (n=20).

Method (canonical). MNE ERDS-maps recipe (Pfurtscheller & Lopes da Silva 1999): per-epoch **multitaper** TFR (freqs 2–35 Hz, `n_cycles = freqs`), **percent-change** baseline (–1→0 s), `decim`=3. Read-outs: ERDS time-frequency maps at C3/Cz/C4 and mu-band (8–13 Hz) topographies over the 0.5–3.5 s task window. Epochs are buffered (-2.0–4.5 s) and cropped to -1.0–3.9 s after baseline to remove multitaper edge ringing. Preprocessing matches EEGPT (average ref, 0–38 Hz, 256 Hz). Convention: **percent < 0 = ERD** (desync).

Result — mu lateralization (group, C3 − C4). - LEFT imagery (T1): **+0.6%** (expect > 0 → contralateral C4 desync) - RIGHT imagery (T2): **-6.7%** (expect < 0 → contralateral C3 desync)

Central sensorimotor mu ERD is present during imagery; the ERDS maps show the mu/beta suppression time course at C3/Cz/C4.

Significance (cluster-permutation, group n=20). Two-sided one-sample cluster test vs baseline per channel/class; outlined regions on the ERDS maps are significant (p<0.05, corrected for the many time-frequency comparisons). Min cluster p — C3 during RIGHT imagery: 0.001; C4 during LEFT imagery: 0.020. Full table in `cluster_stats.csv`.

This is the empirical reference for later model comparison.

Artifacts: `qc.csv`, `erd_mu.csv`, `cluster_stats.csv`, `erds.npz`, figures; `config.json` records exact parameters and package versions.

Per-subject QC

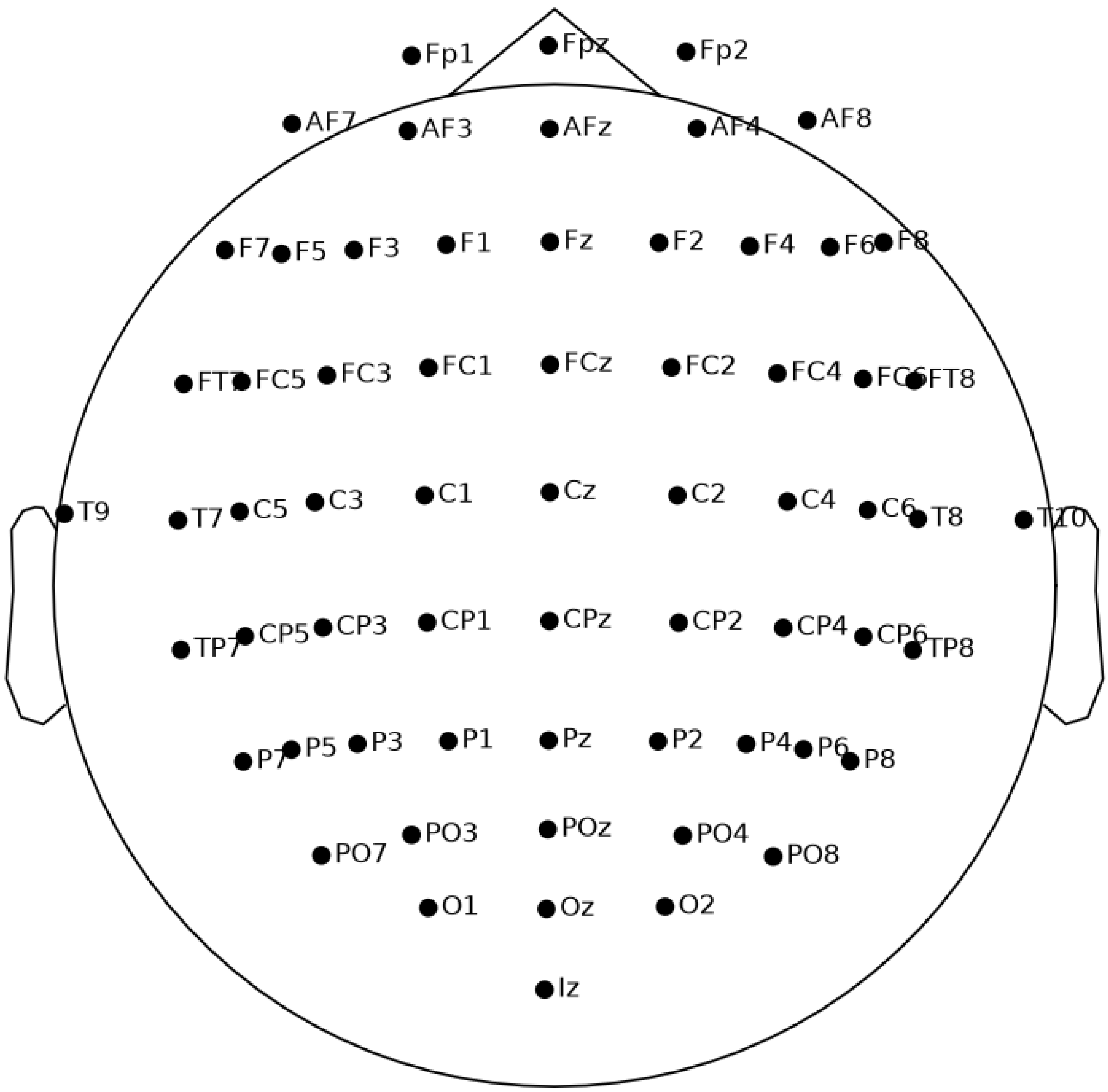
subject	sfreq	n_channels	T1	T2	high_amp_pct	ok
1	256.0	64	23	22	0.33	True
2	256.0	64	23	22	0.07	True
3	256.0	64	23	22	2.53	True
4	256.0	64	23	22	0.99	True
5	256.0	64	21	24	0.05	True
6	256.0	64	24	21	0.04	True
7	256.0	64	23	22	0.28	True
8	256.0	64	22	23	0.29	True
9	256.0	64	24	21	5.52	True
10	256.0	64	24	21	11.7	True
11	256.0	64	23	22	0.06	True
12	256.0	64	21	24	0.9	True
13	256.0	64	23	22	2.49	True
14	256.0	64	22	23	0.13	True
15	256.0	64	23	22	0.47	True
16	256.0	64	22	23	0.31	True
17	256.0	64	23	22	7.77	True
18	256.0	64	22	23	0.0	True
19	256.0	64	23	22	0.67	True
20	256.0	64	23	22	0.0	True

Cluster-permutation significance

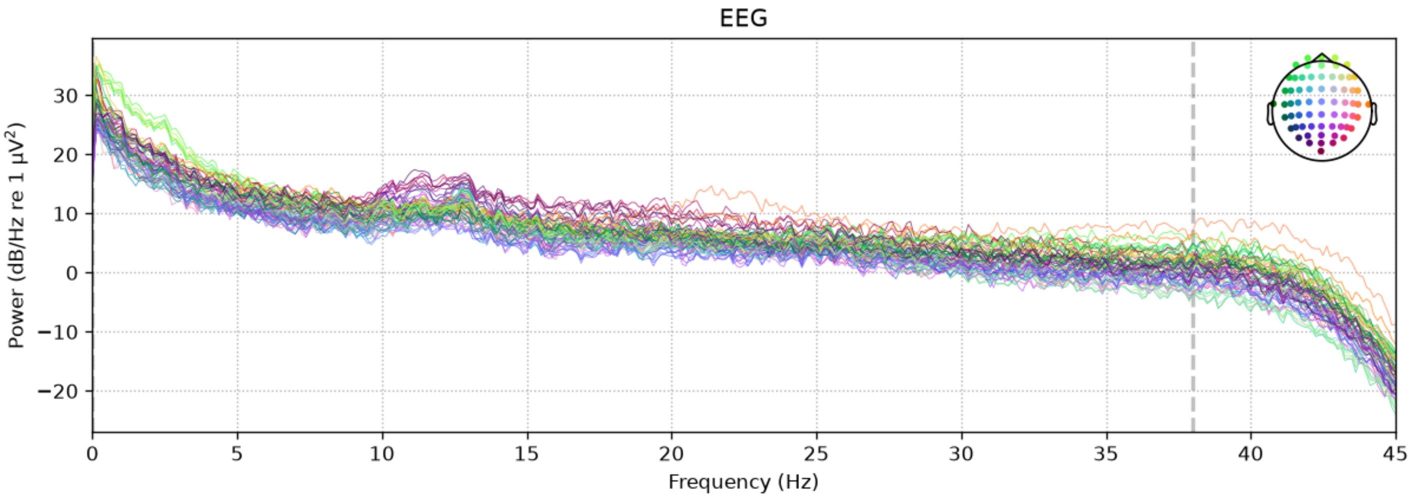
class	channel	min_cluster_p	significant
T1	C3	0.0117	True
T1	Cz	0.0371	True
T1	C4	0.0195	True
T2	C3	0.001	True
T2	Cz	0.0029	True
T2	C4	0.0059	True

Figures

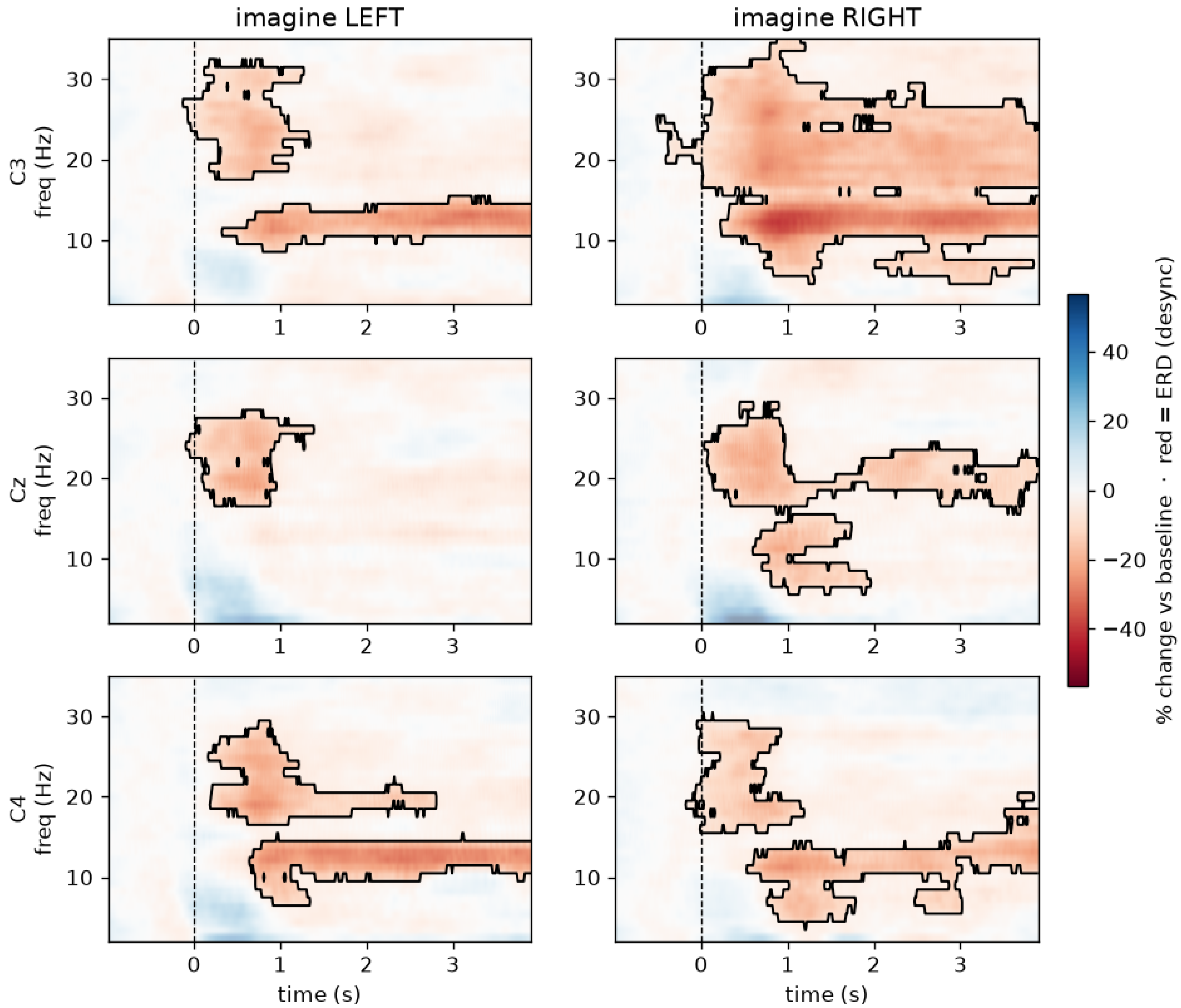
Sensor montage (64-ch, 10-10)



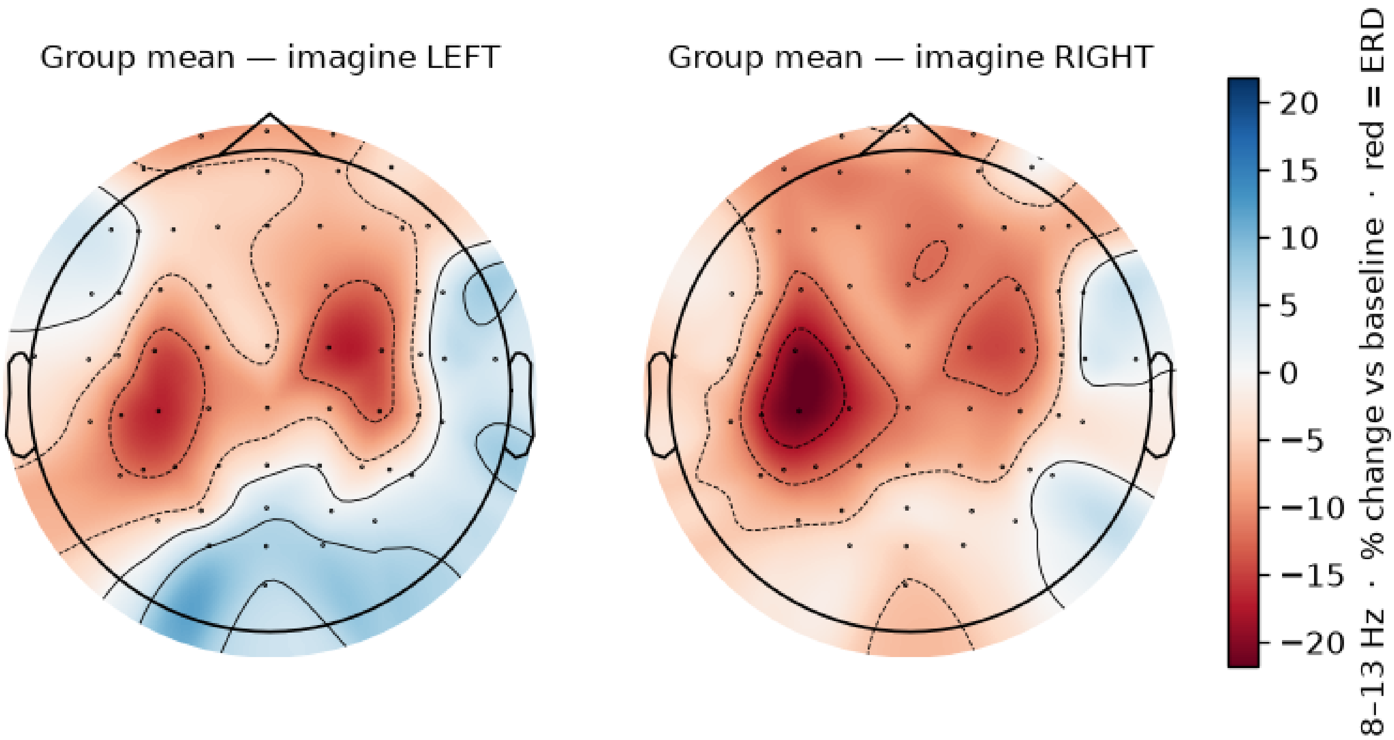
Power spectral density (0–45 Hz)



ERDS maps at C3/Cz/C4 (group; red = ERD)



Mu-band ERD topography — group mean (red = ERD)


Group mean — imagine LEFT
Group mean — imagine RIGHT
20
15
10
5
0
−5
−10
−15
−20
8–13 Hz · % change vs baseline · red = ERD